

External Merge Sort: Cost Analysis

Overview

External Merge Sort is used when a file is too large to fit in main memory. **Main costs:**

1. **Block Transfers** (read/write)
2. **Disk Seeks** (moving the disk head)

1 Total Block Transfers

Block Transfer Formula

$$\text{Total Transfers} = br \times \left(2 \lceil \log_{M-1}(br/M) \rceil + 1 \right)$$

Symbols:

- br : Number of blocks in the file
- M : Number of memory blocks available
- $\lceil \cdot \rceil$: Ceiling function (round up)
- $\log_{M-1}(br/M)$: Number of merge passes

Stepwise Breakdown:

Step 1: Run Generation: Read all blocks, sort in memory, and write back as sorted runs.

$$\text{Cost} = br \text{ (read)} + br \text{ (write)} = 2br$$

Step 2: Merge Passes: Each pass reads and writes all blocks: $2br$ per pass. Number of passes:

$$\lceil \log_{M-1}(br/M) \rceil$$

Step 3: Final Pass Optimization: The last pass can directly output sorted data without writing back to disk, saving br block writes.

Derivation:

$$\text{Total Transfers} = 2br \text{ (run gen)} + 2br \cdot \lceil \log_{M-1}(br/M) \rceil - br = br \times \left(2\lceil \log_{M-1}(br/M) \rceil + 1 \right)$$

Intuition: Every pass reads all blocks; writes happen except in the final pass.

Visual Idea

Run Generation: Read + Write (2br)

Merge Pass 1: Read + Write (2br)

Merge Pass 2: Read + Write (2br)

Final Merge Pass: Read only (br)

2 Total Disk Seeks

Disk Seek Formula

$$\text{Total Seeks} = 2 \left\lceil \frac{br}{M} \right\rceil + \left\lceil \frac{br}{b_b} \right\rceil (2P - 1)$$

Symbols:

- br : Number of blocks in the file
- M : Number of memory blocks
- b_b : Buffer size (blocks read sequentially per seek)
- P : Number of merge passes
- $\lceil \cdot \rceil$: Ceiling function (round up)

Stepwise Breakdown:

Step 1: Run Generation Seeks: Each run requires 1 seek to read + 1 seek to write.

$$\text{Runs} = \left\lceil \frac{br}{M} \right\rceil, \quad \text{Total seeks} = 2 \left\lceil \frac{br}{M} \right\rceil$$

Step 2: Merge Pass Seeks: Each chunk of b_b blocks requires 1 seek.

$$\text{Seeks per pass} = \left\lceil \frac{br}{b_b} \right\rceil$$

- Each pass except the final has read + write seeks = 2 per pass - Final pass only reads $\rightarrow$ total seek factor = $2P - 1$

$$\text{Total merge seeks} = \left\lceil \frac{br}{b_b} \right\rceil (2P - 1)$$

Intuition: Seeks measure disk head movements. Larger buffers or memory reduce seeks. Final pass optimization reduces one set of write seeks.

Visual Idea

Run Generation: 2 seeks per run

Merge Passes: $\lceil br/b_b \rceil$ seeks per pass, total $2P - 1$

Summary Table

Cost Type	Formula	Intuition
Block Transfers	$br (2\lceil \log_{M-1}(br/M) \rceil + 1)$	Reads all blocks in every pass; final pass pipelined
Disk Seeks	$2\lceil br/M \rceil + \lceil br/b_b \rceil (2P - 1)$	Counts disk head movements; larger buffers reduce seeks